MODULE 5A · EPISODE 1

The Earth's Atmosphere: Structures and Layers

From the Weather Below to the Edge of Space

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

The Earth's Atmosphere: Structures and Layers

This episode journeys upward through the thin envelope of gas that makes life on Earth possible. The panel begins with the atmosphere's composition and its four main layers, separated by boundary zones called pauses, where the temperature trend reverses. It then climbs through each layer: the troposphere, where all our weather happens and jet streams blow; the stratosphere, home of the ozone layer; the frigid mesosphere, with its noctilucent clouds, ionosphere, and meteors; and the thermosphere, where the aurorae glow and outer space begins.

The Panel

Name	Role
Dr. Michael Harrison	Host; astrophysicist
Dr. Alessandro Rossi	Guest; cosmologist, European Space Agency
Dr. Amara Johnson	Guest; theoretical physicist, atmospheric physics

Learning Objectives

After this episode you should be able to:

- Describe the atmosphere's composition and explain why its small amount of carbon dioxide matters.
- Name the four main layers of the atmosphere and the pauses that separate them, and explain what happens to the temperature trend at each pause.
- Explain how the height of the troposphere varies with latitude, and why nearly all of Earth's weather occurs there.
- Describe how temperature and pressure change with height in the troposphere, and give typical values at sea level and at the tropopause.
- Describe jet streams, and explain how aircraft, balloons, birds, and thunderstorm clouds relate to the troposphere and tropopause.
- Explain the stratosphere's temperature inversion, the role of the ozone layer, and the history of the ozone hole.
- Describe the phenomena of the stratosphere and mesosphere, including nacreous clouds, blue jets, noctilucent clouds, airglow, red sprites, and meteors.
- Explain what the ionosphere is and how skywave lets shortwave and ham radio signals travel around the world.
- Explain why the thermosphere is so hot yet would not feel hot, how aurorae form, and where space begins.

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QUICK REFERENCE

Key Terms

Term	Definition
Atmosphere	The gaseous envelope that encapsulates the Earth; a dynamic, life-sustaining system that has evolved over billions of years.
Greenhouse effect	The warming of Earth by gases, such as carbon dioxide, that trap heat; it helps regulate Earth's temperature.
Pause	A relatively thin boundary zone between two layers of the atmosphere, where the temperature trend reverses.
Troposphere	The first and lowest layer of the atmosphere, closest to the surface, where virtually all weather occurs.
Tropopause	The top of the troposphere, where the air stops cooling with height.
One atmosphere	The air pressure at sea level: 101,325 pascals, or 14.7 pounds per square inch.
Jet stream	A high-speed, west-to-east flow of wind found just below the tropopause.
Anviling out	The spreading of a thunderstorm cloud into a flat top when it stops rising at the tropopause.
Cumulonimbus cloud	A towering thunderstorm cloud.
Stratosphere	The second layer of the atmosphere, above the tropopause, where temperature rises (or stays the same) with height.
Stratopause	The top of the stratosphere, at about 50 kilometers at mid-latitudes.
Temperature inversion	A region where temperature rises with height instead of falling.
Ozone layer	A layer in the stratosphere, mostly 15 to 35 kilometers up, where ozone absorbs harmful ultraviolet radiation.
Chlorofluorocarbons (CFCs)	Human-made compounds that caused depletion of the ozone layer and are now banned internationally.
Nacreous clouds	Mother-of-pearl clouds: tiny ice crystals in the polar winter stratosphere that shimmer with iridescent colors.
Blue jets	Lightning-related events that shoot upward from the tops of thunderstorms into the stratosphere.
Mesosphere	The third layer of the atmosphere, above the stratopause, which contains the coldest temperatures on or near Earth.
Mesopause	The top of the mesosphere, at about 85 kilometers at mid-latitudes.
Noctilucent clouds	Clouds of tiny ice crystals in the summer mesosphere that glow with reflected sunlight during deep twilight.
Ionosphere	An irregularly shaped region of electrically charged gas, from about 60 to 600 kilometers up.

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Term	Definition
Plasma	A gas of free electrons and positive ions.
Skywave (skip)	The reflection of radio waves from 3 to 30 million hertz off the ionosphere, back to Earth.
Airglow	A faint, often greenish glow given off by the whole night sky, as atoms and molecules energized by daytime sunlight release that energy.
Red sprites	Lightning-related flashes in the mesosphere, high above thunderstorms, at the moment of powerful cloud-to-ground strikes.
Laser guide star	Sodium atoms near the top of the mesosphere, made to glow by a laser, used to correct a telescope's images.
Meteoroid	A typically small chunk of space rock; it glows as a meteor when it plows through the atmosphere.
Thermosphere	The fourth and final true layer of the atmosphere, from the mesopause to the thermopause.
Thermopause	The top of the thermosphere, at a variable but typically maximum height of 600 kilometers.
Exosphere	The extremely thin outermost region, above the thermopause, that fades gradually into space.
Magnetosphere	Earth's protective magnetic field, which traps charged particles from the solar wind.
Aurorae	The northern and southern lights, produced when charged particles collide with air in the thermosphere.
Kármán line	The altitude of 100 kilometers, widely recognized as the beginning of outer space.
Anacoustic zone	The region, beginning at about 160 kilometers, where the air is too thin to carry ordinary sound.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
78%, 21%, 0.93%, 0.04%	Nitrogen, oxygen, argon, and carbon dioxide in the atmosphere, excluding water vapor
18 km (11 mi)	Height of the troposphere at the equator, about 59,000 feet
11 km (7 mi); 8 km	Height of the troposphere at mid-latitudes (about 36,000 feet); over the poles
~75% / 99%	The troposphere's share of Earth's air by mass / of its water vapor
15°C (59°F)	Average air temperature at sea level worldwide

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Value	What it refers to
−57°C (−70°F)	Average temperature at the tropopause at mid-latitudes
101,325 Pa (14.7 psi)	Air pressure at sea level: one atmosphere
28,000–45,000 ft	Typical cruising altitudes of commercial jets at mid-latitudes
30° N; 50–60° N	Latitudes of the subtropical and polar jet streams
80–140 mph	Typical jet stream speeds (129–225 km/h)
~69,000 ft	Hot air balloon altitude record
37,000 ft	Highest-flying bird on record, Rüppell's griffon vulture
35,000–56,000 ft	Height of thunderstorm anvil tops, mid-latitudes to tropics
50 km (31 mi)	The stratopause at mid-latitudes, about 164,000 feet
99%	Share of Earth's air by mass in the troposphere and stratosphere together
−3°C (27°F)	Temperature at the stratopause
15–35 km; 20–25 km	Extent of the ozone layer; height of peak ozone abundance
~1/1,000	Air pressure at the stratopause, compared with sea level
60,000–70,000 ft	Operating altitudes of the U-2 spy plane
85,000 ft	Highest altitude of the SR-71 Blackbird
~136,000 ft	Alan Eustace's record skydive from a balloon
85 km (53 mi)	The mesopause at mid-latitudes, about 280,000 feet
−90°C (−130°F)	Average temperature at the mesopause
below −130°C (−200°F)	Mesopause temperature over the summer poles
60–600 km	Extent of the ionosphere
3–30 MHz	Radio frequencies that skywave works for (3 to 30 million hertz)
600 km (373 mi)	Typical maximum height of the thermopause
2,000°C (3,632°F)	Maximum thermosphere temperature, high sunspot activity
500°C (932°F)	Maximum thermosphere temperature, low sunspot activity
~300 km	Where the ionosphere's electrons are most densely packed
100–500 km	Altitudes where aurorae glow
100 km	The Kármán line, the beginning of outer space
160 km	Where the anacoustic zone begins
400 km (250 mi)	Orbit of the International Space Station

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — Composition and Structure

As far as we know, Earth is the only “rock” in the entire Universe that supports life, and its atmosphere is central to that uniqueness. We can define the atmosphere as the **gaseous envelope that encapsulates the Earth**, but it is much more than a layer of gas: it is a dynamic, life-sustaining system that has evolved over billions of years.

Excluding water vapor, the atmosphere consists primarily of four gases: **78% nitrogen, 21% oxygen, 0.93% argon, and only about 0.04% carbon dioxide**. Small as it is, that carbon dioxide plays a crucial role in regulating Earth's temperature through the **greenhouse effect**.

The atmosphere is divided into **four main layers**, separated by relatively thin boundary zones called **pauses**, where the temperature trend reverses. Each layer has its own characteristics, temperature profile, and phenomena.

Part 2 — The Troposphere

The **troposphere** is the first layer, closest to the surface where we live and breathe. Its height varies with latitude: about **18 kilometers** (11 miles, 59,000 feet) at the equator, about **11 kilometers** (7 miles, 36,000 feet) at mid-latitudes, and only about **8 kilometers** over the poles.

Though it is the thinnest layer, the troposphere is incredibly dense with matter. It holds about **75% of Earth's air by mass** and **99% of the atmosphere's water vapor**, which is why virtually all of Earth's weather occurs there: weather needs water in its various phases.

Temperature **decreases with height** in the troposphere, from an average of **15°C (59°F)** at sea level to about **−57°C (−70°F)** at the top, the **tropopause**, at mid-latitudes. Above the tropopause, the air stops cooling and eventually starts to warm again. Sea-level pressure is **101,325 pascals**, or 14.7 pounds per square inch, which we call **one atmosphere**. Both temperature and pressure fall dramatically with height.

Part 3 — Flying Through the Troposphere

At mid-latitudes, commercial jets cruise at about **28,000 to 45,000 feet**, in the upper troposphere and, on the highest flights, just into the lower stratosphere. The retired **Concorde** cruised at around 60,000 feet, well above the tropopause, avoiding much of the weather and turbulence below.

Just below the tropopause flow the **jet streams**, high-speed winds blowing from west to east. In the Northern Hemisphere, the **subtropical jet stream** lies near 30° N, and the **polar jet stream** wanders at about 50 to 60° N. Their winds typically reach **80 to 140 miles per hour** (129 to 225 km/h), and eastbound airliners ride them as tailwinds to save time and fuel.

Most hot air balloons stay far below the jet streams, though record-setting round-the-world balloon flights have ridden them, and one hot air balloon has reached a record of about **69,000 feet**. The highest-flying bird on record, **Rüppell's griffon vulture**, has been observed at

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37,000 feet over Africa.

Because the temperature trend reverses at the tropopause, towering **cumulonimbus** thunderstorm clouds usually stop rising there and spread into a flat top, from about 35,000 feet at mid-latitudes to about 56,000 feet in the tropics. This is called **“anviling out.”**

Part 4 — The Stratosphere

The **stratosphere** is the second layer, extending at mid-latitudes from the tropopause to about **50 kilometers** (31 miles, 164,000 feet), the **stratopause**. Together, the troposphere and stratosphere hold **99% of Earth's air by mass**.

Counterintuitively, temperature **rises (or stays the same)** with height in the stratosphere. This **temperature inversion** comes from interactions between ozone, oxygen, and the Sun's high-frequency ultraviolet radiation. The stratopause can reach a comparatively balmy **−3°C (27°F)**.

The **ozone layer** lies mostly between **15 and 35 kilometers**, with peak abundance at 20 to 25 kilometers. Along with oxygen, it absorbs the most destructive wavelengths of the Sun's ultraviolet radiation, which would cause serious DNA damage if it reached the surface. Human release of **chlorofluorocarbons (CFCs)** produced an ozone hole over Antarctica; an international ban has let the layer slowly start healing. Near the surface, ozone is a pollutant that can impair lung function.

In the polar winter, water vapor can freeze into tiny ice crystals that form **nacreous**, or mother-of-pearl, **clouds**, which shimmer during civil twilight. **Blue jets** shoot upward from thunderstorm tops into the stratosphere. At the stratopause, the air pressure is only about **one thousandth** of sea-level pressure, far too thin for jet engines. The **U-2** spy plane operated at 60,000 to 70,000 feet and the **SR-71 Blackbird** flew as high as 85,000 feet, while **Alan Eustace** sky-dived from a helium-filled balloon at just under 136,000 feet.

Part 5 — The Mesosphere

The **mesosphere** is the third layer, extending at mid-latitudes from the stratopause to about **85 kilometers** (53 miles, 280,000 feet), the **mesopause**. Temperature **decreases** with height, making this the coldest region on or near Earth: about **−90°C (−130°F)** at the mesopause, and below **−130°C (−200°F)** over the summer poles, colder than anywhere on Earth's surface.

There, water vapor, some of it from the breakdown of methane, freezes into tiny ice crystals that form **noctilucent clouds**. Strangely, they appear in summer, glowing with reflected sunlight during deep twilight, long after sunset.

The **ionosphere**, an irregular region of electrically charged gas, begins in the mesosphere at about 60 kilometers and extends to about 600 kilometers. Solar radiation knocks electrons loose from atoms and molecules, creating a **plasma** of free electrons and positive ions that can reflect radio waves. This reflection, **skywave** or **skip**, works for frequencies from **3 to 30 million hertz** and lets shortwave and ham radio operators communicate around the world.

On moonless nights, the whole sky gives off a faint, often greenish **airglow**, as atoms and molecules energized by daytime sunlight slowly release that energy as light. **Red sprites** flash

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high above thunderstorms at the moment of powerful cloud-to-ground lightning strikes.

Near the top of the mesosphere, at about 90 kilometers, a thin layer of **sodium atoms** left by burned-up meteors can be made to glow with a laser. These **laser guide stars** let telescopes continuously adjust flexible mirrors to remove the blurring caused by atmospheric turbulence. And it is in the mesosphere that **meteoroids** burn up: plowing through the thin air at tremendous speeds, they compress and heat the air so intensely that they glow.

Part 6 — The Thermosphere

The **thermosphere** is the fourth and final true layer. It starts at the mesopause and extends to a variable but typically maximum height of **600 kilometers** (373 miles), the **thermopause**. Above it, the extremely thin **exosphere** fades gradually into space.

As in the stratosphere, temperature **rises** with height in the lower thermosphere, and the maximum depends dramatically on sunspot activity: about **2,000°C (3,632°F)** on the sunlit side during high activity, and still about **500°C (932°F)** during low activity. Yet because the air is so thin, you wouldn't actually feel hot there.

The ionosphere's electrons are most densely packed at about **300 kilometers**. Throughout the thermosphere, the few air molecules absorb the Sun's very short-wave ultraviolet and X-ray radiation, which ionizes them and gives them tremendous kinetic energy, producing those extreme temperatures.

The thermosphere is home to the **aurorae**. The **magnetosphere** traps charged particles from the solar wind and funnels some down along magnetic field lines into oval zones around the north and south magnetic poles, where they collide with air **100 to 500 kilometers** up. Green, the most common color, and the deep red seen highest both come from **oxygen** atoms; the blue, violet, and pinkish-red fringe at the bottom comes from **nitrogen** molecules: a giant neon sign in the sky.

The thermosphere also marks boundaries for space exploration. At **100 kilometers**, the **Kármán line**, the atmosphere is so thin that it is widely recognized, including by the European Space Agency, as the beginning of outer space. At about **160 kilometers**, the **anacoustic zone** begins, where ordinary sound can no longer travel. And the **International Space Station** orbits within the thermosphere at about **400 kilometers**, or 250 miles.

From the weather-generating troposphere, through the ozone-rich stratosphere and the frigid mesosphere, to the thermosphere where space begins, each layer plays its part in making Earth the remarkable oasis of life it is.

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STUDY QUESTIONS

Questions with Model Answers

Question numbering matches the student study guide exactly. Model answers are written at the level expected of a strong student response, not as the only acceptable wording.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Composition and Structure

1. What makes Earth unique in the Universe, as far as we know? What role does its atmosphere play?

ANSWER As far as we know, Earth is the only “rock” in the entire Universe that supports life. Its atmosphere is absolutely central to this uniqueness.

2. How can we define Earth's current atmosphere? Why is it “so much more than just a layer of gas”?

ANSWER It is the gaseous envelope that encapsulates the Earth. It is more than a layer of gas because it is a dynamic, life-sustaining system that has evolved over billions of years.

3. Excluding water vapor, which four gases make up most of Earth's atmosphere, and in what proportions?

ANSWER About 78% nitrogen, 21% oxygen, 0.93% argon, and only about 0.04% carbon dioxide.

4. Carbon dioxide makes up only about 0.04% of the atmosphere. Why is it nevertheless important?

ANSWER This small percentage plays a crucial role in regulating Earth's temperature through the greenhouse effect.

5. How many main layers is the atmosphere divided into? What are the “pauses,” and what happens at them?

ANSWER Four main layers. The pauses are relatively thin boundary zones between the layers, where the temperature trend reverses.

Part 2 — The Troposphere

1. Where is the troposphere? How does its height vary with latitude? Give its height at the equator, at mid-latitudes, and over the poles.

ANSWER It is the first layer, closest to the surface. Its top is about 18 km (11 miles, 59,000 feet) high at the equator, about 11 km (7 miles, 36,000 feet) at mid-latitudes, and only about 8 km over the poles.

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2. The troposphere is the thinnest layer, yet it is described as incredibly dense with matter. What shares of Earth's air and water vapor does it contain?

ANSWER About 75% of Earth's air by mass and about 99% of the atmosphere's water vapor.

3. Why does virtually all of Earth's weather occur in the troposphere?

ANSWER Because it holds almost all of the atmosphere's water vapor, and you need water in its various phases to create weather systems.

4. How does air temperature change with height in the troposphere? Give the average temperature at sea level and at the tropopause at mid-latitudes.

ANSWER It decreases with increasing height, from an average of about 15°C (59°F) at sea level to about −57°C (−70°F) at the tropopause.

5. What is the air pressure at sea level, and what do we call this pressure? What happens to pressure as you climb?

ANSWER 101,325 pascals, or 14.7 pounds per square inch, which we call one atmosphere. Pressure decreases dramatically with height.

6. What happens to the temperature trend above the tropopause?

ANSWER The air stops cooling and eventually starts to warm again.

Part 3 — Flying Through the Troposphere

1. At what altitudes do commercial jets typically cruise at mid-latitudes? How was the Concorde an exception, and what did it gain?

ANSWER About 28,000 to 45,000 feet, in the upper troposphere and, on the highest flights, the lower stratosphere. The Concorde cruised at around 60,000 feet, well above the tropopause, which let it avoid much of the weather and turbulence of the troposphere.

2. What are jet streams, and where are they found? Name the two jet streams of the Northern Hemisphere and give their latitudes.

ANSWER High-speed west-to-east flows of wind found just below the tropopause, in both hemispheres. The subtropical jet stream lies near 30° N, and the polar jet stream wanders at about 50 to 60° N.

3. How fast do jet stream winds move? Why do eastbound airliners like to “ride” them?

ANSWER Typically 80 to 140 miles per hour (129 to 225 km/h), sometimes faster. Riding them as tailwinds reduces travel time and saves fuel.

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4. How do hot air balloons relate to the jet streams? What is the hot air balloon altitude record?

ANSWER Most hot air balloons stay far below the jet streams, but record-setting round-the-world balloon flights have deliberately ridden them. The altitude record for a hot air balloon is about 69,000 feet.

5. What is the highest-flying bird on record? How high has it been observed, and in which layer?

ANSWER Rüppell's griffon vulture, observed at 37,000 feet, high in the troposphere over Africa.

6. Why do thunderstorm clouds usually stop rising at the tropopause? What is “anviling out,” and at what heights does it happen?

ANSWER Because the temperature trend reverses there, cumulonimbus clouds stop rising and spread out into a flat top. This is anviling out, and it happens from about 35,000 feet at mid-latitudes to about 56,000 feet in the tropics.

Part 4 — The Stratosphere

1. Where is the stratosphere, and how high does it extend at mid-latitudes? What is its upper boundary called?

ANSWER It is the second layer, just above the tropopause. At mid-latitudes it extends to about 50 km (31 miles, 164,000 feet). Its upper boundary is the stratopause.

2. What share of Earth's air by mass do the troposphere and stratosphere hold together?

ANSWER 99%.

3. How does temperature change with height in the stratosphere, and what causes this? How warm can it get at the stratopause?

ANSWER It rises (or stays the same) with increasing height, the opposite of the troposphere. This temperature inversion is due to interactions between ozone, oxygen, and the Sun's high-frequency ultraviolet radiation. The stratopause can reach about −3°C (27°F), close to freezing.

4. Where is the ozone layer, and where is ozone most abundant? What does the ozone layer protect life from?

ANSWER Mostly between about 15 and 35 km, with peak abundance at about 20 to 25 km. Along with oxygen, it absorbs the most destructive wavelengths of the Sun's ultraviolet radiation, which would cause serious DNA damage if it reached the surface.

5. What caused the ozone hole over Antarctica, and what has allowed the ozone layer to begin healing? Where is ozone considered a pollutant?

ANSWER Humanity's release of chlorofluorocarbons (CFCs). The international ban on these compounds has let the layer slowly start healing. Near Earth's surface, ozone is a pollutant that can seriously impair lung function.

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6. What are nacreous clouds? When and where do they form, and when can they be seen?

ANSWER Mother-of-pearl clouds made of extremely small ice crystals. They form in winter in the polar stratosphere, when temperatures drop low enough for water vapor to freeze, and shimmer with iridescent colors during civil twilight.

7. What are blue jets?

ANSWER Lightning-related events that shoot upward from the tops of thunderstorms into the stratosphere.

8. How low is the air pressure at the stratopause? Give the altitudes reached by the U-2, the SR-71 Blackbird, and Alan Eustace.

ANSWER About one thousandth of sea-level pressure, far too thin for any air-breathing jet engine. The U-2 operated at 60,000 to 70,000 feet, the SR-71 flew as high as 85,000 feet, and Alan Eustace sky-dived from a helium balloon at just under 136,000 feet.

Part 5 — The Mesosphere

1. Where is the mesosphere, and how high does it extend at mid-latitudes? What is its upper boundary called?

ANSWER It is the third layer, just above the stratopause. At mid-latitudes it extends to about 85 km (53 miles, 280,000 feet). Its upper boundary is the mesopause.

2. Why is the mesosphere remarkable for its temperatures? Give the average mesopause temperature and the coldest temperatures over the summer poles.

ANSWER It contains the coldest temperatures on or near Earth, because temperature decreases with height. The mesopause averages about −90°C (−130°F), and over the summer poles it can plunge below −130°C (−200°F), colder than anywhere on Earth's surface.

3. What are noctilucent clouds? What are they made of, when do they appear, and when can they be seen?

ANSWER Clouds of extremely small ice crystals, formed from water vapor, some of it from the breakdown of methane. Strangely, they appear in summer, glowing with reflected sunlight during deep twilight, long after sunset.

4. What is the ionosphere? Where does it begin and end, and how does it form?

ANSWER An irregularly shaped region of electrically charged gas that begins in the mesosphere at about 60 km and extends to about 600 km. Solar radiation knocks electrons loose from atoms and molecules, creating a plasma of free electrons and positive ions.

5. What is skywave, or skip? Which radio frequencies does it work for, and who uses it?

ANSWER The reflection of radio waves from the ionosphere back to Earth. It works for frequencies from 3 million to about 30 million hertz, and shortwave radio enthusiasts and ham radio operators use it to communicate around the world.

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6. What is airglow, and what causes it?

ANSWER A faint glow, often greenish, given off by the whole sky on moonless nights far from city lights. Atoms and molecules high in the atmosphere, energized by sunlight during the day, slowly recombine and release that energy as light.

7. What are red sprites?

ANSWER Lightning-related events in the mesosphere that flash high above thunderstorms at the same moment as powerful cloud-to-ground lightning strikes below.

8. What is a laser guide star, and how do astronomers use it?

ANSWER A spot of sodium atoms, near the top of the mesosphere at about 90 km, made to glow like an artificial star by a finely tuned laser. Telescopes use it to continuously adjust flexible mirrors, removing the distortions caused by atmospheric turbulence.

9. What are meteoroids, and why do they glow as they pass through the mesosphere?

ANSWER Typically small chunks of space rock. Plowing through the thin air at tremendous speeds, they compress and heat up so intensely that they glow.

Part 6 — The Thermosphere

1. Where does the thermosphere begin and end? What lies above it?

ANSWER It starts at the mesopause and extends to a variable but typically maximum height of 600 km (373 miles), the thermopause. Above it, the extremely thin exosphere fades gradually into space.

2. How do temperatures in the thermosphere depend on sunspot activity? Give the maximum temperatures during high and low activity.

ANSWER They rise with height in the lower thermosphere, and the maximum depends dramatically on sunspot activity: about 2,000°C (3,632°F) on the sunlit side during high activity, and about 500°C (932°F) during low activity.

3. If the thermosphere is so hot, why wouldn't you feel hot there?

ANSWER Because the air is so thin. There are too few molecules to transfer much heat to you.

4. What causes the thermosphere's extreme temperatures? Where are the ionosphere's electrons most densely packed?

ANSWER The few air molecules absorb very short-wave ultraviolet and X-ray radiation from the Sun, which ionizes them and gives them tremendous kinetic energy. The electrons are most densely packed at about 300 km.

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5. How are aurorae produced? Where on Earth, and at what altitudes, do they occur?

ANSWER The magnetosphere traps charged particles from the solar wind and funnels some of them down along magnetic field lines into oval zones around the north and south magnetic poles. They collide with air in the thermosphere at about 100 to 500 km, creating the northern and southern lights.

6. Which gases produce the different auroral colors?

ANSWER Green, the most common color, and the deep red at the highest altitudes both come from excited oxygen atoms. The blue, violet, and pinkish-red fringe at the bottom of some displays comes from nitrogen molecules.

7. What is the Kármán line, and at what altitude is it? What begins at about 160 km?

ANSWER At 100 km, the atmosphere is so thin that this altitude is widely recognized, including by the European Space Agency, as the beginning of outer space. At about 160 km, the anacoustic zone begins, where ordinary sound can no longer travel.

8. At what altitude does the International Space Station orbit, and in which layer?

ANSWER About 400 km, or 250 miles, within the thermosphere.

9. For each of the four main layers, state whether temperature rises or falls with height, and name one phenomenon found there.

ANSWER Troposphere: falls; weather, jet streams, anvil clouds. Stratosphere: rises (or stays the same); the ozone layer, nacreous clouds, blue jets. Mesosphere: falls; noctilucent clouds, red sprites, meteors, laser guide stars. Thermosphere: rises; aurorae, the Kármán line, the International Space Station.

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TEACHER NOTES

Corrections and Cautions

Points in the episode that need a qualification, or a prepared answer for the question a sharp student will ask.

C1 **“About 75% of Earth's air by mass” depends on latitude.**

The fraction of air below the tropopause is 1 − (tropopause pressure ÷ sea-level pressure). At mid-latitudes (226 mb) that is about 78%; in the tropics, where the tropopause is far higher (about 100 mb), it is about 90%. Textbooks usually quote 75 to 80% as a global figure. The Module 5A layers slideshow shows 78% for mid-latitudes and 90% for the tropics.

C2 **Carbon dioxide is rising.**

0.04% is 400 parts per million; the 2025 average was about 425 ppm, up from about 280 ppm before the Industrial Revolution. The percentages exclude water vapor because it varies from nearly 0 to about 4% depending on place and weather.

C3 **Layer heights and temperatures are averages.**

The pauses move with latitude, season, and weather; the episode's heights are mid-latitude values. Published temperatures also vary: the mid-latitude mesopause is often quoted between −85 and −100°C, so some of the course graphics show −148°F (−100°C) rather than the episode's −130°F (−90°C). Both are within the normal range, and the summer polar mesopause can fall to about −140°C.

C4 **Why the temperature trend reverses.**

Ozone and oxygen absorb ultraviolet light and heat the stratosphere from above, so it warms with height. The lower stratosphere is nearly constant in temperature up to about 20 km, which is why the episode says temperatures “rise (or stay the same).” The mesosphere, with little ozone, cools with height again; the thermosphere is heated by the far shorter ultraviolet and X-rays absorbed in Part 6.

C5 **Jet streams wander.**

The polar jet can range from about 40 to 60° N and shifts with the seasons, and peak winds can exceed 200 mph. Westbound flights try to avoid it: a Chicago–London flight is often more than an hour shorter than the return.

C6 **The balloon records.**

Vijaypat Singhania set the hot air balloon altitude record of 68,986 feet over India in 2005. The round-the-world flights, Breitling Orbiter 3 (1999) and Steve Fossett's Spirit of Freedom (2002), used Rozière balloons, which combine helium and hot air. Alan Eustace's 2014 jump, from 135,890 feet, rode a helium balloon and broke Felix Baumgartner's 2012 record.

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C7 **The vulture's 37,000 feet.**

This record comes from a 1973 collision between a Rüppell's griffon vulture and an airliner over Abidjan, Côte d'Ivoire. The species has a special form of hemoglobin that grips oxygen in thin air.

C8 **Anvils and overshooting tops.**

The anvil spreads where the rising air reaches the warmer, stable stratosphere. The strongest storms briefly punch an “overshooting top” into the stratosphere above the anvil, a sign of severe weather.

C9 **The ozone layer is recovering slowly.**

The 1987 Montreal Protocol phased out CFCs. The latest UN assessment expects the Antarctic ozone layer to return to 1980 levels around 2066. Nacreous (polar stratospheric) clouds are part of the story: chemistry on their ice surfaces releases the chlorine that destroys ozone each spring. Ground-level ozone is a different problem, formed in smog from vehicle and industrial emissions: “good up high, bad nearby.”

C10 **The spy planes.**

The U-2 first flew in 1955 and is still in service. The SR-71 made its last flight in 1999; its record altitude was 85,069 feet, set in 1976.

C11 **Meteors, not meteor showers.**

Each glowing streak is a *meteor*; a *meteor shower* is many meteors on one night, when Earth crosses a comet's debris trail. Most meteors glow at about 70 to 120 km, so they span the upper mesosphere and lower thermosphere. The glow comes mainly from ram compression of the air in front of the meteoroid, not from friction.

C12 **Red sprites and blue jets.**

Sprites appear a few thousandths of a second after strong positive cloud-to-ground strikes and span about 50 to 90 km. They were first photographed in 1989; blue jets were first recorded from aircraft in 1994. Both are hard to see with the unaided eye.

C13 **The ionosphere and skywave.**

Many sources extend the ionosphere to about 1,000 km. Its D, E, and F regions change from day to night, and at night AM radio (0.5 to 1.7 MHz) also travels by skywave, which is why distant AM stations come in after dark.

C14 **“You wouldn't feel hot.”**

Temperature measures the average speed of the molecules; heat transfer depends on how many of them there are. The thermosphere is so thin that a thermometer would barely warm. An astronaut there would be heated by direct sunlight and chilled in shadow, which is why spacesuits and spacecraft need careful thermal control.

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C15 **The Kármán line is a convention.**

Theodore von Kármán estimated that, near this height, an aircraft would need orbital speed to stay aloft. The 100 km line is used by the FAI and ESA, while the U.S. Air Force, NASA, and FAA award astronaut wings at 50 miles (about 80 km). Some researchers argue that 80 km is the better boundary.

C16 **Aurora colors and the ISS.**

The green comes from oxygen at 557.7 nanometers, mostly at 100 to 250 km; the red from oxygen at 630 nanometers, above about 250 km. The Space Station orbits at roughly 400 to 420 km at about 17,500 mph, and thermospheric drag slowly lowers its orbit, so it is periodically reboosted.

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TEACHER NOTES

Teaching the Episode

Build the scale model.

Run the basketball scale model from the Try It box, or use a 1-meter stick for 100 km: then the ISS is 4 m away, and the troposphere is only the first 11 cm. Students are usually shocked at how thin the life-supporting layer is.

Plot the temperature profile.

Give students the episode's temperatures at sea level, the tropopause, the stratopause, the mesopause, and the thermosphere, and have them graph temperature against height. The zigzag makes the four layers and the pauses obvious, and the reversals become the definition of a pause.

Show what pressure does.

Seal an empty plastic bottle at a high altitude (or on a flight) and watch it crush on the way down. In class, a marshmallow in a sealed syringe swells as you pull the plunger. Connect both to the fall from one atmosphere to a thousandth of it at the stratopause.

Track the jet streams.

Open a live wind map, such as earth.nullschool.net, set to the 250 mb level, and find both jet streams. Then compare scheduled flight times between two cities in each direction: the eastbound trip is shorter.

Test the ozone shield.

UV-sensitive beads change color in sunlight. Have students test them behind window glass, sunscreen, and sunglasses, then discuss what the ozone layer does on a planetary scale.

Look up.

Check the next major meteor shower and an International Space Station pass using NASA's Spot the Station. Students who see the ISS are watching a spacecraft move through the thermosphere.

Use the course graphics.

The jet stream, ozone layer, ionosphere, aurora, thermosphere, and edge-of-space slides line up with Parts 3 to 6. Ask students to explain each one in a sentence before you show the narration.

Use the panel.

Alessandro, a cosmologist from the European Space Agency, opens with Earth as the only known life-bearing “rock,” and the episode ends with ESA's Kármán line. Ask students why someone who studies the whole Universe would care about a layer of gas only a few hundred kilometers thick.

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